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Labor-Market Search and International Business Cycles

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The objective of this paper is to explain the observed international fluctuations by modifying the traditional modelling of the labor market in the two-country real business cycles model. Our intuition is that labor-market search can be useful to understand the propagation of international fluctuations. Changes in the expected returns to search induce responses in search and recruiting activities, the effects of which are propagated through time via changes in the stock of employment. Given that the technology shock spills over to the other country, domestic and foreign firms start searching at the same time in anticipation. Employment then increases simultaneously and displays a hump-shaped profile in the two countries. This partially curtails the capital outflow from the country which does not benefit from the shock. Introducing in a search framework a non-separability between consumption and leisure in the utility function allows both to solve the consumption puzzle and to complete the explanation of the international comovement puzzle. *Journal of Economic Literature* Classification Numbers: E32, F41, J40.

Key Words: international business cycles; search; wage bargaining.

1. INTRODUCTION

International real business cycles (RBC) models have been extensively used to explain the observed aggregate correlations across countries. International fluctuations of consumption and output have received the main attention since the seminal work of Backus *et al.* (1992). Faced with the inability of the canonical one-good two-country model to replicate empirical

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features (Backus *et al.,* 1992), according to which cross-country correlations of output are higher than the one of consumption (the so-called consumption correlation puzzle), numerous extensions have been proposed in the literature. Devereux *et al.* (1992), for instance, show that the non-separability of consumption and leisure in the utility function can generate a realistic correlation between national consumptions. Stockman and Tesar (1995) also succeed in replicating this correlation by incorporating non-traded goods and taste shocks. Several recent studies choose to modify the constraints on trade among agents: Baxter and Crucini (1995) and Heathcote and Perri (1999) build models with incomplete asset markets which reduce the incentive for risk sharing. However, there is another puzzle, the so-called international comovement puzzle (Baxter, 1995): all the international real business cycles models tend to predict substantial, negative international comovement of labor inputs and investments, whereas the observed correlations are strongly positive.

> "A major challenge to the theory is to develop a model which can explain international comovement in labor input and investment" Baxter [1995], p. 1859.

A model with multiple goods and important linkages between goods (either on the consumption or the production side) could potentially generate realistic international comovements of labor and investment. However, the two-good model (Backus *et al.,* 1994) cannot counter the effects of the productivity differential between production factors across countries arising from technology shocks for realistic elasticity of substitution between the two goods.[2] This flaw leads to negative international correlations of employment and investment, which arise from the fact that capital flows are the primary channel of international transmission. The overriding force behind investment dynamics in this two-good model is still at work. In the traditional two-country optimal growth model, the employment dynamics cannot reverse the influence of capital. It even reinforces the negative international transmission: a positive technology country-specific shock creates a mutual positive wealth effect, which induces a decrease of the labor supply in the other country.

The aim of this paper is to explain the observed international fluctuations simply by modifying the traditional modelling of the labor market in the two-country RBC model. Our intuition is that labor-market search can be useful to understand the propagation of international fluctuations arising from productivity shocks. Since Merz (1995) and Andolfatto (1996), labor-market search in the lines of Pissarides (1990) appeared as a convenient and useful way to introduce non-walrasian features in a RBC framework, giving micro-foundations to unemployment in a general equilibrium setting.

[2]Ambler *et al.* (1998) are more successful by incorporating multiple tradable goods sector.

In a closed economy perspective, this model is consistent not only with labor-market stylized facts such as the persistence in unemployment and the asymmetric dynamic correlation between hours and labor productivity, but also with business cycles features such as the degree of output autocorrelation. It must be underlined that we only consider technology shocks, whereas the international comovement puzzle has been recently partially solved by taking into account monetary shocks in a pricing-to-market and nominal rigidities framework (Betts and Devereux, 2000; Chari *et al.,* 1998) or home production shocks (Canova and Ubide, 1998).

We show in this paper that labor-market search can also help in understanding international business cycles.[3] Because finding new workers takes time and effort, firms view their existing workforce as a capital asset. Changes in the expected returns to search induce responses in search and recruiting activities, the effects of which are propagated through time via changes in the stock of employment. Employment then cannot react abruptly to a shock, as in the standard model. Given the Solow residuals bivariate process, implying international diffusion, a technology shock arising in one country gives incentives for firms of the other country to post more vacancies. Employment rises slowly in the two countries, displaying a typical hump-shaped response, due to the time-consuming nature of the search process. Higher employment may also curtail the capital outflow from the second country by increasing capital productivity.

Instead of systematically studying the sensitivity of the search model empirical outcomes to the calibration, the method chosen consists of comparing the search two-country model with the walrasian one, by the means of stochastic simulations and impulse response functions for a reasonable calibration. In a first step, we compare a "pure" search model to a "pure" walrasian one in order to explain and quantitatively evaluate the intrinsic propagation mechanisms of the search two-country model; we then reveal that they come closer to the observed international comovements. In a second step, we show that introducing in a search framework a non-separability between consumption and leisure in the utility function, as already done by Devereux *et al.* (1992) in a two-country framework, allows to solve both the consumption puzzle and the international comovement puzzle. By eliminating the wealth effect in the bargaining process and thanks to a strong internal propagation mechanism, the search model empirical outcome then becomes very close to the data.

Section 2 is devoted to the presentation of the model, while the calibration and the simulation results of the benchmark model are presented

[3]Shi (1999) has already used this framework in an international context to analyze the implications of tariffs on unemployment.

in Section 3. The implications of a non-separable utility function are examined in Section 4.

2. THE MODEL

Each country specializes in the production of a single good affected by technology shocks A. Households consume a CES basket of the two goods indexed by 1 (domestic good) and 2 (foreign good). They participate in the trade on the labor market. Employees work h units of time at a wage rate of w and unemployed workers devote e units of their time to seeking employment.

2.1. The Search Process

Let $N_{i,t}$ denote the number of jobs that are matched with a worker at the beginning of the period t in the country i. As the population size is normalized to 1, $U_{i,t}$ is the measure of the unemployment rate.[4] Let $V_{i,t}$ be the total number of new jobs made available by firms during the period t, each vacancy incurring a flow cost equal to ω. Exit rates from unemployment and vacancy are determined by the search and recruiting decisions of workers and firms; these decisions enter as complementary inputs into an aggregate matching function. The rate $H_{i,t}$ at which new job matches are formed is governed by an aggregate matching function:

$$H_{i,t} = \chi V_{i,t}^{\phi}(e_i U_{i,t})^{1-\phi}, \qquad \text{for } i = 1, 2.$$

χ is a measure of the efficiency of the matching process. e_i is the exogenous search effort per worker seeking employment, and $e_i U_i$ is the aggregate search effort. Employment then evolves according to

$$N_{i,t+1} = (1-s)N_{i,t} + H_{i,t}, \qquad \text{for } i = 1, 2,$$

where s is the exogenous rate of separation. Hence a job vacancy can at best become productive only after one period of time has elapsed.

2.2. Households

Each household participates in the labor market and consumes a CES basket C_i^c of the two goods indexed by 1 (domestic good) and 2 (foreign good),

$$C_i^c = \left[\gamma^{1/\theta} C_i^{(\theta-1)/\theta} + (1-\gamma)^{1/\theta} C_{j\neq i}^{(\theta-1)/\theta}\right]^{\theta/(\theta-1)}, \qquad \text{for } i = 1, 2,$$

[4]Following Andolfatto (1996), the distinction between unemployed and not in the labor force is ignored here.

where θ is the elasticity of substitution between the two goods. We can define a price index in each country,

$$P_i^c = \left[\gamma P_i^{1-\theta} + (1-\gamma) P_{j\neq i}^{1-\theta}\right]^{1/(1-\theta)}, \qquad \text{for } i = 1, 2,$$

where P_i is the price of the good i. We normalize the price of the domestic good (good 1) to 1 and denote by P the (relative) price of the foreign good (good 2). This latter corresponds to the terms of trade for the foreign country: the relative price of exported goods to imported goods. An increase of P will correspond to an appreciation (depreciation) of the terms of trade for the foreign country (domestic country).

The random matching and separation processes induce different employment histories among households between employed workers (n) and unemployed workers (u), implying potential wealth heterogeneity. Assuming perfect insurance market, following Hansen (1985), there will be, however, no ex-post heterogeneity in accumulation choices. We can derive the intertemporal decisions rules by solving a representative household program.

We assume perfect international risk sharing: households have access to contingent claims B at prices v, providing one unit of good 1 if the state A occurs (we denote by $f(A)$ the density function associated to the stochastic variable A). The country 1 representative household is assumed to maximize the expected discounted sum of its utility flows,

$$W_1^H(B_{1,t}) = \max\left\{N_{1,t}U_{1,t}^n + (1-N_{1,t})U_{1,t}^u + \beta\int W_1^H(B_{1,t+1})f(A)\,dA\right\},$$

subject to the following constraints,

$$N_{1,t}P_{1,t}^c C_{1,t}^{c,n} + (1-N_{1,t})P_{1,t}^c C_{1,t}^{c,u} + \int v_t B_{1,t+1}\,dA \leq B_{1,t} + N_{1,t}w_{1,t}h_{1,t}, \qquad (\lambda_{1,t})$$

$$N_{1,t+1} = (1-s)N_{1,t} + t_{1,t}e_1(1-N_{1,t}). \qquad (\zeta_{1,t})$$

The country 2 analogue is

$$W_2^H(B_{2,t}) = \max\left\{N_{2,t}U_{2,t}^n + (1-N_{2,t})U_{2,t}^u + \beta\int W_2^H(B_{2,t+1})f(A)\,dA\right\},$$

subject to the following constraints,

$$\begin{aligned} &N_{2,t}P_{2,t}^c C_{2,t}^{c,n} + (1-N_{2,t})P_{2,t}^c C_{2,t}^{c,u} + \int v_t B_{2,t+1}\,dA \\ &\quad \leq B_{2,t} + N_{2,t}P_t w_{2,t}h_{2,t}, \qquad (\lambda_{2,t}) \end{aligned}$$

$$N_{2,t+1} = (1-s)\,N_{2,t} + t_{2,t}e_2(1-N_{2,t}), \qquad (\zeta_{2,t})$$

with $t_{i,t} = H_{i,t}/e_i U_{i,t}$. We note that the wage rate w_i is defined in terms of the good produced at home.

According to Andolfatto (1996), the preferences are given by the following instantaneous utility function,[5]

$$U_i^s = \log C_i^{c,s} + \Gamma_s(h_i^s), \qquad \text{for } s = n, u \quad \text{and} \quad i = 1, 2,$$

with $\Gamma_n(h_i) = \sigma_n[(1 - h_i)^{1-\eta}/(1 - \eta)]$ and $\Gamma_u(e_i) = \sigma_e[(1 - e_i)^{1-\eta}/(1 - \eta)]$. h_i denotes hours per worker and e denotes the time devoted to search. The parameter σ_s can be interpreted as reflecting differences in the efficiency of household's home production technology across different states of employment opportunities. Under perfect insurance market, one gets

$$\begin{aligned} C_{i,t}^{c,n} &= C_{i,t}^{c,u}, \\ U_i^u - U_i^n &= \Gamma_u(e_i) - \Gamma_n(h_{i,t}). \end{aligned} \tag{1}$$

The optimality conditions are

$$\frac{1}{C_{i,t}} = P_{i,t}^c \lambda_{i,t}, \tag{2}$$

$$v_t = \beta \frac{\lambda_{i,t+1}}{\lambda_{i,t}} f(A). \tag{3}$$

2.3. Firms

The two countries produce imperfectly substitutable goods with capital and labor: each country specializes in the production of a single good. The goods are produced with constant return to scale technology,

$$Y_i = A_i K_i^{c1-\alpha}(h_i N_i)^\alpha, \qquad \text{for } i = 1, 2.$$

We assume that investment to each country is an index of the two goods with the same structure as the consumption one. The capital index K^c evolves according to the following dynamic equation:

$$K_{i,t+1}^c = (1 - \delta)K_{i,t}^c + I_{i,t}^c.$$

We assume there are capital adjustment costs $\mathscr{C}_{i,t}$ (Obstfeld and Rogoff, 1996):

$$\mathscr{C}_{i,t} = \frac{\Phi}{2} \frac{(K_{i,t+1}^c - K_{i,t}^c)^2}{K_{i,t}^c}.$$

[5] We will introduce later on a non-separability between consumption and leisure in the utility function.

Fluctuations arise from persistent shocks to aggregate productivity,

$$\begin{pmatrix} \log A_{1,t} \\ \log A_{2,t} \end{pmatrix} = \begin{pmatrix} \rho & \rho_* \\ \rho_* & \rho \end{pmatrix} \begin{pmatrix} \log A_{1,t-1} \\ \log A_{2,t-1} \end{pmatrix} + \begin{pmatrix} (1-\rho) & -\rho_* \\ -\rho_* & (1-\rho) \end{pmatrix} \times \begin{pmatrix} \log A_1 \\ \log A_2 \end{pmatrix} + \begin{pmatrix} 1 & \psi \\ \psi & 1 \end{pmatrix} \begin{pmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \end{pmatrix}, \tag{4}$$

where A_1 and A_2 are the means of the processes followed by $A_{1,t}$ and $A_{2,t}$, respectively. $\varepsilon_{1,t}$ and $\varepsilon_{2,t}$, which represent the innovations to productivity variables, share the same variance σ_A^2 and are such that $E(\varepsilon_{1,t}) = E(\varepsilon_{2,t}) = 0$, $E(\varepsilon_{1,t}\varepsilon'_{2,t}) = 0$. ψ is a positive parameter which determines the contemporaneous correlation between domestic and foreign technology shocks.

The program of each firm in country 1 is dynamic and consists of maximizing the expected discounted sum of profit flows,

$$W_1^F(K_{1,t}^c, N_{1,t}) = \max\Big\{ Y_{1,t} - \mathscr{C}_{1,t} - P_{1,t}^c I_{1,t}^c - \omega_1 V_{1,t} - w_{1,t} h_{1,t} N_{1,t} + \int v_t W_1^F(K_{1,t+1}^c, N_{1,t+1})\, dA \Big\},$$

subject to the following constraints,

$$K_{1,t+1}^c = (1-\delta)K_{1,t}^c + I_{1,t}^c, \tag{$q_{1,t}$}$$

$$N_{1,t+1} = (1-s)N_{1,t} + m_{1,t}V_{1,t}, \tag{$\xi_{1,t}$}$$

with $m_{1,t} = H_{1,t}/V_{1,t}$. The optimality conditions are, using (3),

$$\xi_{1,t} m_{1,t} = \omega_1, \tag{5}$$

$$\xi_{1,t} = \beta E_t \left\{ \frac{\lambda_{1,t+1}}{\lambda_{1,t}} \left(\frac{\partial Y_{1,t+1}}{\partial N_{1,t+1}} - w_{1,t+1} h_{1,t+1} + \xi_{1,t+1}(1-s) \right) \right\}, \tag{6}$$

$$q_{1,t} = P_{1,t}^c + \frac{\partial \mathscr{C}_{1,t}}{\partial I_{1,t}}, \tag{7}$$

$$q_{1,t} = \beta E_t \left\{ \frac{\lambda_{1,t+1}}{\lambda_{1,t}} \left(\frac{\partial Y_{1,t+1}}{\partial K_{1,t+1}} - \delta P_{1,t+1}^c + q_{1,t+1} \right) \right\}. \tag{8}$$

The program of each firm in country 2 is analogous,

$$W_2^F(K_{2,t}^c, N_{2,t}) = \max\Big\{ P_t[Y_{2,t} - \mathscr{C}_{2,t} - \omega_2 V_{2,t} - w_{2,t} h_{2,t} N_{2,t}] - P_{2,t}^c I_{2,t}^c + \int v_t W_2^F(K_{2,t+1}^c, N_{2,t+1})\, dA \Big\},$$

subject to the following constraints,

$$K^c_{2,t+1} = (1-\delta)K^c_{2,t} + I^c_{2,t}, \qquad (q_{2,t})$$

$$N_{2,t+1} = (1-s)N_{2,t} + m_{2,t}V_{2,t}, \qquad (\xi_{2,t})$$

with $m_{2,t} = H_{2,t}/V_{2,t}$. The optimality conditions are, using (3),

$$\xi_{2,t} m_{2,t} = P_t \omega_2, \tag{9}$$

$$\xi_{2,t} = \beta E_t \left\{ \frac{\lambda_{2,t+1}}{\lambda_{2,t}} \left(P_{t+1} \frac{\partial Y_{2,t+1}}{\partial N_{2,t+1}} - P_{t+1} w_{2,t+1} h_{2,t+1} + \xi_{2,t+1}(1-s) \right) \right\}, \tag{10}$$

$$q_{2,t} = P^c_{2,t} + P_t \frac{\partial \mathscr{C}_{2,t}}{\partial I_{2,t}}, \tag{11}$$

$$q_{2,t} = \beta E_t \left\{ \frac{\lambda_{2,t+1}}{\lambda_{2,t}} \left(P_{t+1} \frac{\partial Y_{2,t+1}}{\partial K_{2,t+1}} - \delta P^c_{2,t+1} + q_{2,t+1} \right) \right\}. \tag{12}$$

The labor decisions ((5), (6), (9), and (10)) depart from the ones gotten in the traditional two-country two-good RBC model. They are similar to the capital ones ((7), (8), (11), and (12)). Because finding new workers takes time and effort, firms view their existing workforce as a capital asset. Another important implication of our approach comes from the fact that wages and individual hours are bargained. The surplus generated by a match is shared between firms and households according to their bargaining power.

2.4. Wage and Hours Bargaining

Following Pissarides (1990), the bargaining process is given by the solution to the Nash criterion,

$$\max_{w,h} (\lambda \Omega^F)^{\epsilon} (\Omega^H)^{1-\epsilon},$$

with $\Omega^F = \partial W^F/\partial N$ the marginal value of a match for a firm and $\Omega^M = \partial W^H/\partial N$ the marginal value of a match for a worker. ϵ denotes the firm's share of a job's value. We deduce from this program the bargained wage and hours contract,

$$w_{1,t} h_{1,t} = (1-\epsilon) \left\{ \frac{\partial Y_{1,t}}{\partial N_{1,t}} + \mu_{1,t}\omega_1 \right\} + \epsilon \left\{ \frac{\Gamma_u(e_1) - \Gamma_n(h_{1,t})}{\lambda_{1,t}} \right\}, \tag{13}$$

$$\alpha \frac{Y_{1,t}}{N_{1,t} h_{1,t}} (1-h_{1,t})^{\eta} = \frac{\sigma_n}{\lambda_{1,t}}, \tag{14}$$

for country 1 and

$$w_{2,t}h_{2,t} = (1-\epsilon)\left\{\frac{\partial Y_{2,t}}{\partial N_{2,t}} + \mu_{2,t}\omega_2\right\} + \epsilon\left\{\frac{\Gamma_u(e_2) - \Gamma_n(h_{2,t})}{P_2\lambda_{2,t}}\right\}, \quad (15)$$

$$\alpha\frac{Y_{2,t}}{N_{2,t}h_{2,t}}(1-h_{2,t})^{\eta} = \frac{\sigma_n}{P_t\lambda_{2,t}}, \quad (16)$$

with $\mu_i = V_i/eU_i$. The worker's wage bill is a weighted average of its contribution to output and its outside opportunities. The firm bargaining power ϵ determines the spread between the labor productivity and the wage bill, which explains vacancy decisions ((6) and (10)). A decrease of λ_i increases outside opportunities, giving an incentive to employees for working less and claiming higher wages. Induced by a positive productivity shock in country 1, for instance, this effect tends to increase the product wage in country 2. On the contrary, this shock improves the terms of trade in country 2, leading to a decrease in its product wage. This provides country 2 firms a larger share of the match surplus, which not only increases their incentive for posting more vacancies, but may also curtail the capital outflow from country 2.

3. QUANTITATIVE EVALUATION

Since we cannot analytically find the equilibrium, log-linearized approximations to the equilibrium decision rules around the steady state are computed for this economy using the method proposed by King *et al.* (1988). We consider a null net exports steady state. An artificial economy is then calibrated and the equilibrium is computed numerically. We apply the standard treatment of data for the computation of the statistics in the real business cycle literature. The moments reported in the different tables are computed from Hodrick–Prescott filtered artificial time series.

3.1. Calibration of the Artificial Economy

The procedure for calibrating is traditional: we choose share parameters for preferences and production such that means of ratios of aggregate times series are equal to analogous ratios for the theoretical economy's steady state. For numerous parameters, we use the same calibration as in Backus *et al.* (1994) and in Andolfatto (1996). We adopt a symmetrical calibration between the two countries.

The technology parameters are calibrated according to Backus *et al.* (1994). See Table I. The cross-country diffusion parameter ρ^* is fixed at 0.088 and ψ is calibrated in order to get a correlation between technology innovations of 0.258. The depreciation rate δ is set at 0.025. It must be

TABLE I
Technology Parameters

δ	Φ	α	ρ	ρ^*	ψ	σ_A
0.025	1.63	0.65	0.906	0.088	0.133	0.00852

pointed out that α no longer corresponds to the labor share of output. We calibrate it in order to get a labor share of 64%. Φ, the capital adjustment cost parameter, is calibrated in order to replicate the volatility of investment in our "realistic economy" (Table VI). We keep it constant across the other experiments in order to isolate the intrinsic properties of the different models.

Following Backus *et al.* (1994), θ, the elasticity of substitution between domestic and foreign goods, is set at 1.5, while the discount factor β is set at 0.99. The value of γ allows to replicate the 15% stationary ratio of imports to GDP. See Table II.

For labor-market parameters, the calibration is based on Andolfatto (1996). The values of σ_n and σ_u are chosen in order to get stationary values of 0.57 for N, the employment ratio, and of 1/3 for h, the average fraction of time spent working. Aggregate expenditures on search activity are fixed at 1% of GDP and the average fraction of time that nonemployed households devote to search is set to $\frac{1}{2}h$. χ is fixed to get a value of 0.9 for the probability that a vacant position becomes a productive job. ϕ is at the value 0.60, estimated by Blanchard and Diamond (1989). We set ϵ at the same value, implying that Hosios's (1990) condition is satisfied: following Merz (1995) and Andolfatto (1996), wage and hours bargaining is such that trade externalities do not distort equilibrium. The rate of transition from employment to nonemployment is set at 0.10, according to Davis *et al.* (1996). η is set such that the individual labor supply elasticity is in the range of the values reported by MaCurdy (1981). See Table III.

3.2. Intrinsic Business Cycles Properties

The equilibrium decision rules are used to simulate the time paths for the variables of interest. The statistical properties of these simulated time

TABLE II
Preferences Parameters

β	η	θ	γ
0.99	5	1.5	0.85

TABLE III
Labor Market Parameters

$\frac{\omega V}{Y}$	χ	ϕ	s	ϵ
1%	0.25	0.60	0.10	0.60

series are then compared to the statistical properties of the corresponding data coming from Backus *et al.* (1995).

We aim at checking that our search model can enhance the traditional explanation of the international comovement. At this stage we focus mainly on input comovements across countries (Table IV). We will turn to the consumption puzzle in the next section (Table V). In order to compare with a traditional international RBC model, we also simulate a two-good two-country model, but without search on the labor market, corresponding then to a Backus *et al.* (1994) style model.[6] The results of these simulations are reported in the line "walrasian labor market." We adopt the same calibration strategy for the walrasian model in order to isolate the effects stemming from the particular search hypothesis.[7] We follow here the same approach as in Andolfatto (1996): the value added by introducing labor-market search is assessed by comparing its performance relative to a benchmark RBC model. As we want to have at hand a walrasian model which is nested within the search model, we consider a finite intertemporal substitution elasticity of leisure model:[8] we thus focus on total hours correlation comparison, since employment is constant in the walrasian framework.[9]

[6]Our model without search (walrasian labor market) does not correspond exactly to Backus *et al.*'s (1994) model, in particular because there is capital adjustment cost instead of time-to-build. Moreover, one may notice that, as Backus *et al.* (1994) are more interested by the price puzzle, they do not report the investment and employment correlations across countries. We verify that we obtain output and consumption correlations across countries which are very close to the ones reported in Backus *et al.* (1994, Table 4, p. 100).

[7]The parameters δ, Φ, ρ, ρ^*, ψ, σ_A, β, η, θ, γ take the same value, while α is equal to 0.64. The parameters reported in Table III do not appear in the walrasian two-good two-country model.

[8]Another strategy would be to only consider employment dynamics in the search model and then compare with a walrasian one including the Hansen (1985) indivisible labor hypothesis. We have verified that it does not change our conclusion.

[9]This strategy could be problematic because the data on total hours are available for very few countries. However, employment and total hours correlations across countries are very close for those countries for which total hours are measured, as evidenced by Amber *et al.* (1999). Hereafter, we will consider that the stylized facts on employment cross-correlation documented by Backus *et al.* (1995) can be used for discriminating between walrasian and search models on the basis of total hours dynamics. An alternative strategy would have been to consider the stylized facts on total hours recently presented by Ambler *et al.* (1999), instead of the traditional ones. The conclusion of this paper would not have been altered.

TABLE IV
Cross-country Business Cycles

	Output	Labor	Investment	Consumption
Data (median)	0.51	0.36	0.38	0.40
Search labor market	0.29	0.25	0.08	0.71
Walrasian labor market	0.25	−0.18	−0.19	0.80

The first line of Table IV gives the median correlation of each macro aggregate with the corresponding U.S. aggregate (Backus *et al.*, 1995). Outputs of developed countries appear to move together. This statement also applies for consumption, labor, and investment. The correlation is larger for output than for consumption, the latter being roughly similar to that of investment and labor. The inability of traditional international RBC models to reproduce these features has been much documented, especially the international input comovements, when only technology shocks are taken into account (Backus *et al.*, 1995; Baxter, 1995). The two-country search model appears more in accordance with the empirical correlations, even if it does not perfectly succeed in replicating the observed magnitude of international comovements. As can be seen in Table IV, introducing search leads to a positive comovement for total hours and a non-negative one for investment, whereas we get negative cross-correlations in the walrasian case. These results derive from the particular employment behavior in the search model, whereby the dynamics are indeed roughly similar for employment and total hours. The employment cross-correlation is equal to 0.30, which is very close to that of total hours.[10] Individual hours are indeed weakly affected, as can be seen in Fig. 2 by inspecting the instantaneous response of total hours. This indicates that the employment dynamics in a search context explains the differences in the two models' outcomes.

The simulation results can be explained by computing the impulse response functions following a productivity shock in country 1 (domestic country). The key mechanism for understanding positive aggregate comovements refers to the search process on the labor market. Given that the technology shock spills over to the other country, domestic and foreign firms start searching at the same time in anticipation (Fig. 1).

Total hours rise slowly in the two countries (Fig. 2), displaying a typical hump-shaped response. When we compare the responses of labor and investment in the labor-market search economy relative to the walrasian one, it appears that this particular hump-shaped profile response in the former model explains why we partially solve the international comovement puzzle. The total hours dynamics are indeed totally different in the

[10]This result is in accordance with the facts recently presented by Ambler *et al.* (1999).

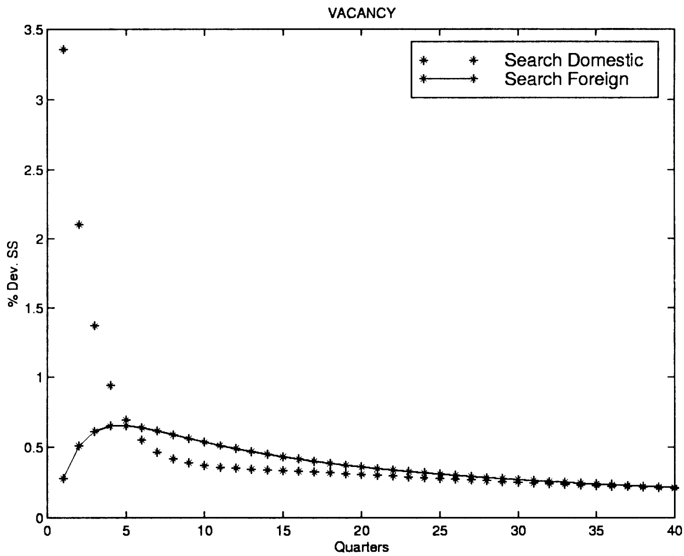

FIG. 1. Responses to a domestic shock (benchmark model).

walrasian model, even if the instantaneous responses are similar: domestic hours monotonously decrease afterwards, whereas the foreign ones increase in the walrasian economy. The hump-shaped feature in the labor-market search has already been used by Andolfatto (1996) to match the positive

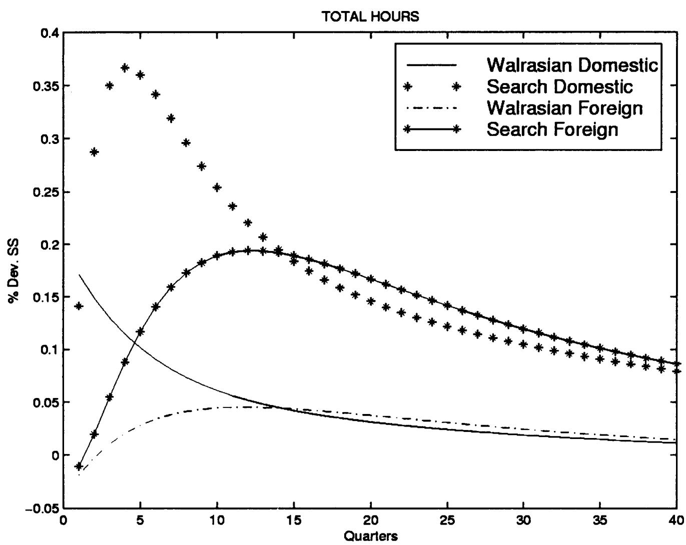

FIG. 2. Responses to a domestic shock (benchmark model).

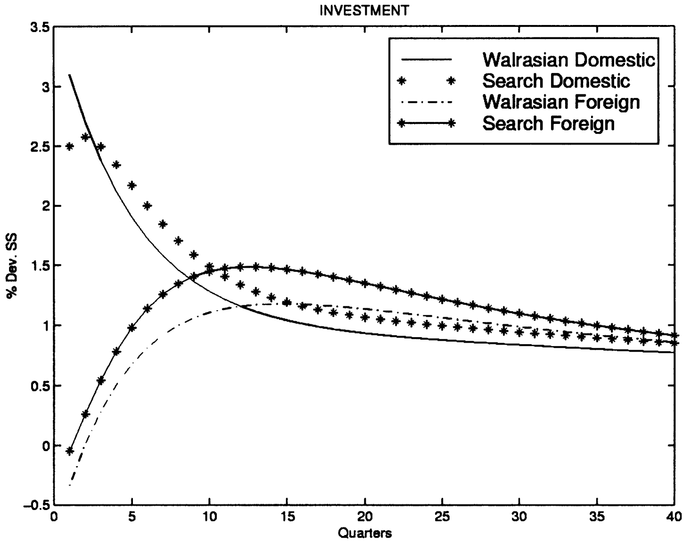

FIG. 3. Responses to a domestic shock (benchmark model).

autocorrelations of output growth rate (Fig. 6 in Appendix). It comes from the time-consuming nature of search, which looks like particular costs of adjustment to the labor input. This strong internal propagation induces positive comovement between total hours across countries.

This particular dynamics interacts with the capital investment, which exhibits a hump-shaped dynamics, too (Fig. 3), leading to improve the replication of the observed investment correlations across countries. Higher employment partially curtails the capital outflow from the second country by increasing the capital productivity. Foreign capital remains almost unchanged in country 2 during the first periods and increases afterwards (Fig. 3).

The results about output correlations are not as good. The correlation is only slightly increased by the labor-market search hypothesis. This is due to the fact that the output dynamics is driven mainly by the exogenous international productivity process in the short run, since individual hours are weakly affected (Fig. 6 in Appendix); the output correlation then is not significantly different between the two models. One may note that this is also the case for the labor productivity correlation, which is equal to 0.30 in both walrasian and search economies. The particular labor dynamics in the search model leads to moderate the increase of the labor productivity, but in both countries (Fig. 7 in Appendix).

The main drawback concerns the consumption correlation: the benchmark labor-market search model remains inconsistent with the observed consumption correlation across countries. This result was expected since there is no mechanism inside the model to disentangle international consumptions (Fig. 8 in Appendix). The next section is partially devoted to this question.

4. A NON-SEPARABLE UTILITY FUNCTION

Up to this point, we can conclude that the labor-market search mechanisms allow one to come closer to the observed international comovement features than does a walrasian model. Now, our aim is to provide a more satisfying explanation of international business cycles by introducing a traditional extension of a two-country RBC model in the search framework. Following Devereux *et al.* (1992), it is well known that the consumption puzzle may be solved by taking into account preferences exhibiting a non-separability between consumption and labor supply:

$$U_i^s = \log(C_i^{c,s} + \Gamma_s(h_i^s)), \qquad \text{for } s = n, u \quad \text{and} \quad i = 1, 2.$$

Moreover, Rogerson and Wright (1988) show that this kind of specification is particularly relevant for the analysis of models with unemployment. Indeed, as soon as unemployed and employed people are considered together, the question of their relative situation has to be tackled. In any model where risk is allocated efficiently, separability between consumption and leisure implies that all agents receive the same income. As unemployed people benefit from higher leisure, they are better off.[11] This is indeed the case in the benchmark model (see (1)). Beyond solving the consumption puzzle, this preference specification can then improve the replication of international comovements. The unemployed agent situation, which determines the outside opportunity of workers, is indeed the threat point of the

[11] Even if risk sharing is assumed, for eliminating the complexities generated by heterogeneity, Rogerson and Wright (1988) have proposed this preference specification allowing to get ex-post situation where employed people are no longer worse off compared to unemployed agents:

$$U^s = \log(C^{c,s} + \Gamma_s(h^s)) + aC^{c,s}, \qquad \text{for } s = n, u.$$

Adding in the utility function a linear term in consumption implies a positive gap between employed and unemployed utility,

$$C_t^{c,n} - C_t^{c,u} = \Gamma_u(e) - \Gamma_n(h_t),$$

$$U^n - U^u = a(\Gamma_u(e) - \Gamma_n(h_t)).$$

Assuming that $\Gamma_u(e) - \Gamma_n(h_t) > 0$, the instantaneous utility of the unemployed worker is less than that of the employed one. The Devereux *et al.* (1992) specification corresponds to the case $a = 0$.

TABLE V
Cross-country Business Cycles (Non-Separable Utility)

	Output	Labor	Investment	Consumption
Data (median)	0.51	0.36	0.38	0.40
Search labor market	0.44	0.61	0.60	0.30
Walrasian labor market	0.31	0.40	−0.24	0.77

latter in the negotiation process. The Devereux *et al.* (1992) specification corresponds to the case where agents have the same utility level.

We then study the quantitative implications of introducing this utility function both in the labor-market search model and in the walrasian one. The calibration is done in exactly the same way as previously. Whereas the walrasian model has still a lot of deficiencies, the labor-market search succeeds in overcoming most of them: the output and consumption correlations are now correctly ordered and there is an international comovement in labor input and investment (Table V).

A common feature to these two frameworks is the initial response of foreign individual hours after a positive domestic technology shock. As the utility specification eliminates wealth effects in household labor decisions, the term of trade effect, which is inherent in our two-good framework, now dominates and induces a positive response (Fig. 4). This effect increases,

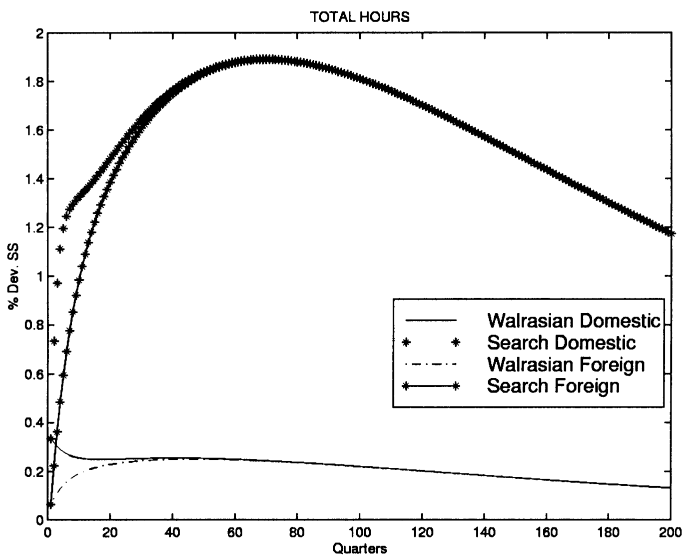

FIG. 4. Responses to a domestic shock (non-separable utility).

ceteris paribus, the total hours correlations, whatever the model considered. In the walrasian case, it mainly explains why the dynamics of leisure is not able to disentangle the dynamics of consumptions, by contrast with the result obtained by Devereux *et al.* (1992) in a one-good model and with the one we get in the search model. The disconnecting effect is indeed much stronger in this latter model (Fig. 4, Fig. 9 in Appendix, and Table V); as a consequence, the consumption correlation across countries deeply declines.

Regarding the international comovement puzzle, the employment dynamics profile in the labor-market search still accounts for the labor and investment international correlations. As has been already mentioned above, the hump-shaped responses of employment and investment in the two countries (Figs. 4 and 5) are crucial to understanding the strong positive correlations across countries concerning investment and total hours. Hiring takes time and employment increases slowly. There are then positive spillover effects on investment dynamics, implying a strong positive correlation of investment across countries. As displayed in Fig. 5, the instantaneous response of the foreign investment is now positive. A closer look at the labor and investment correlations reveals that they are even too high. These input dynamics in the search model lead to a much stronger output correlation across countries (Fig. 10), now superior to the consumption one. Introducing a non-separability between consumption and leisure in the utility function finally tends to exacerbate the discrepancy between the search and the

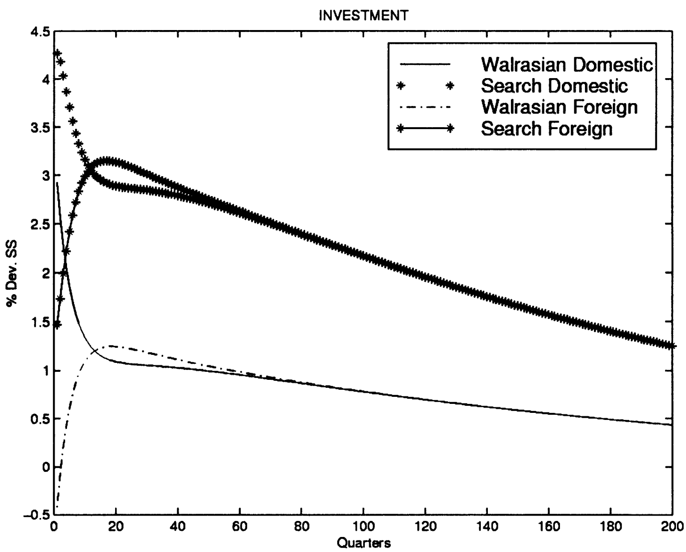

FIG. 5. Responses to a domestic shock (non-separable utility).

TABLE VI
Within-country Business Cycles

	U.S. economy $\sigma_Y = 1.59$		Search economy $\sigma_Y = 1.67$		Walrasian economy $\sigma_Y = 1.29$	
Variable X	(1)	(2)	(1)	(2)	(1)	(2)
Labor	0.93	0.78	0.70	0.90	0.27	0.99
Investment	3.14	0.90	3	0.92	2.39	0.95
Consumption	0.56	0.74	0.57	0.68	0.63	0.95
Labor Productivity	0.64	0.43	0.50	0.80	0.72	0.99

Notes: σ_Y is the percentage standard deviation in output. Column (1) is the standard deviation of the variable X relative to output one. Column (2) is the correlation between X and Y.

walrasian economies.[12] It may be noticed that only the labor productivity correlations across countries are not altered by this preference specification modification: they are still both around 0.30 (Fig. 11).

Of course, it would be a hollow victory to have gotten improved performance at the international level if the search model performs badly on within-country business cycles. It must be actually emphasized that the search model enhances the empirical relevance of the international RBC model along this dimension (Table VI). More particularly, as most of the variance in total hours is accounted for by adjustments in employment rather than hours per worker, the search economy is able to replicate the observed volatility differential between total hours and labor productivity. The output volatility is amplified by the presence of frictions on labor market; this result is in good accordance with the data, too. These results are in line with Andolfatto (1996), so we refrain from a further discussion of these different points.

The labor-market search hypothesis has met with success in capturing many of the salient features of international business cycles, yet the so-called price puzzle remains. Our model is not able to account for the large observed volatility of the terms of trade (Table VII). The negative correlation between output and net exports and between the terms of trade and net exports[13] can be reproduced, but the large volatility of the terms

[12]So far we have considered the same capital adjustment costs scale ($\Phi = 1.63$), the one replicating the investment volatility in the search economy. The results would be even more in favor of the search economy if we had imposed capital adjustment costs so that this volatility is replicated in each economy. As Φ must be decreased in the walrasian one, we get cross-country correlations of output, total hours, investment, and consumption, which are respectively equal to 0.31, 0.35, −0.40, and 0.81, deteriorating the walrasian economy outcome.

[13]We could say that there is a positive correlation between the net exports and the terms of trade defined as the relative price of exports to imports, but we keep the convention used in Backus *et al.* (1994).

TABLE VII
Terms of Trade (P) and Net Exports (NE) Dynamics

	Correlation		Rel. St. Dev.	
Economy	(Y, NE)	(P, NE)	(P)	(NE)
Data (median)	−0.29	−0.46	1.90	0.69
Search labor market	−0.49	−0.89	0.35	0.10
Walrasian labor market	−0.54	−0.72	0.39	0.11

of trade, and even the lower volatility of net exports, are not replicated. Our model misses badly along this dimension.

5. CONCLUSION

Building a two-good two-country labor-market search model allowed us to enlarge the scope of international RBC models. We are able to replicate the positive correlations between aggregates across countries better than does the standard walrasian model. This result derives from the particular internal propagation of technology shocks in a search model. As firms expect that the technology shock spills over to the other country, domestic and foreign firms start searching at the same time. Employment then increases simultaneously and the dynamics displays a hump-shaped profile in the two countries. The time-consuming nature of search is responsible for this last feature. The dynamics of employment partially curtails the capital outflow from the country which does not benefit from the shock. Introducing in a search framework a non-separability between consumption and leisure in the utility function, already used by Devereux *et al.* (1992) in a two-country framework, allows one both to solve the consumption puzzle and to complete the explanation of the international comovement puzzle. By eliminating wealth effect in the bargaining process, and thanks to the strong internal propagation mechanism described above, the search model empirical outcome then becomes very close to the data.

However, the explanation of the great volatility of the terms of trade is left unexplained. Recent papers (Betts and Devereux, 2000; Chari *et al.*, 1998) show that incorporating imperfect competition on the good market and nominal shocks can be considered as essential in the international cycles phenomenon. Combining both frictions on labor and good markets could be fruitful in reaching a more complete explanation of international fluctuations.

APPENDIX

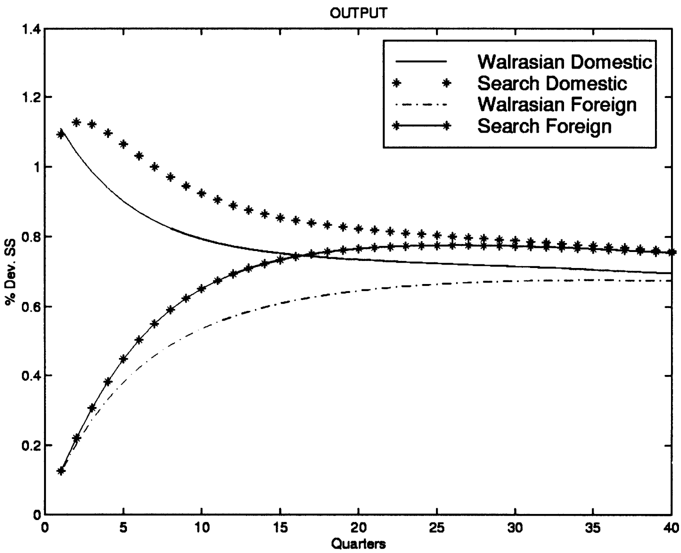

FIG. 6. Responses to a domestic shock (benchmark model).

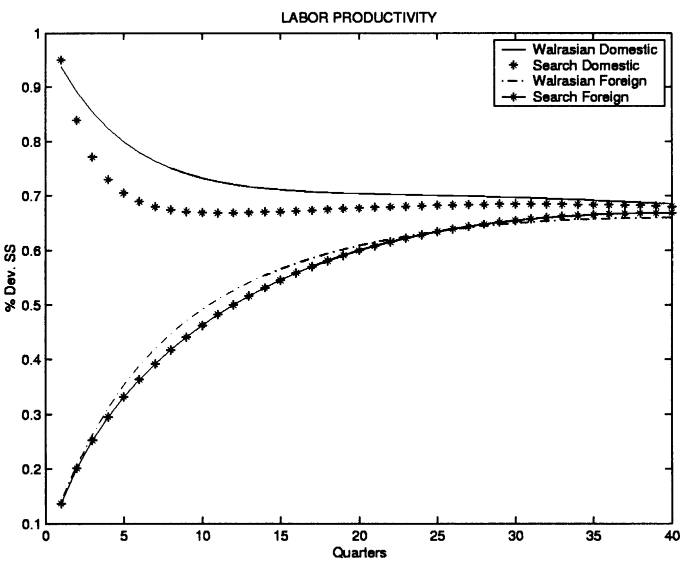

FIG. 7. Responses to a domestic shock (benchmark model).

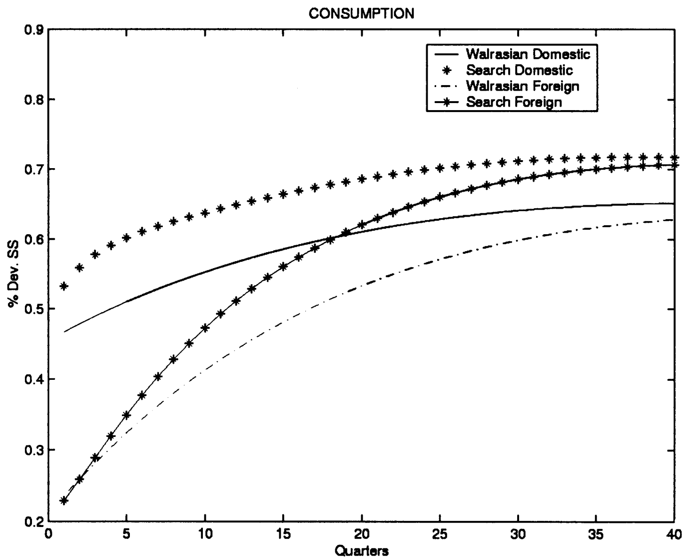

FIG. 8. Responses to a domestic shock (benchmark model).

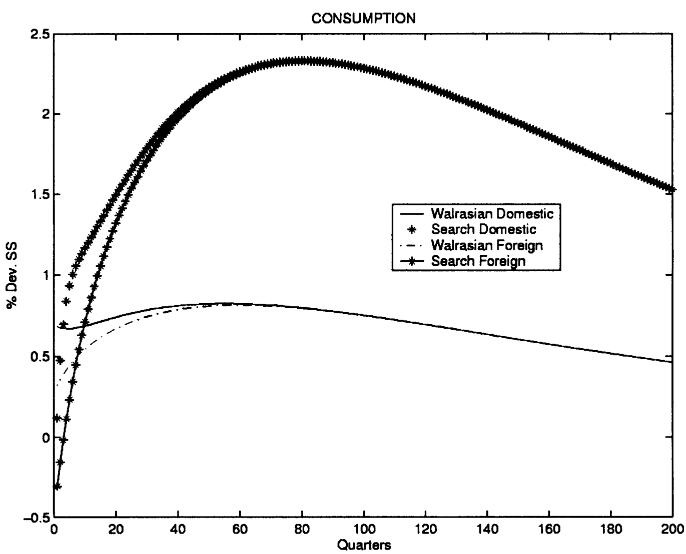

FIG. 9. Responses to a domestic shock (non-separable utility).

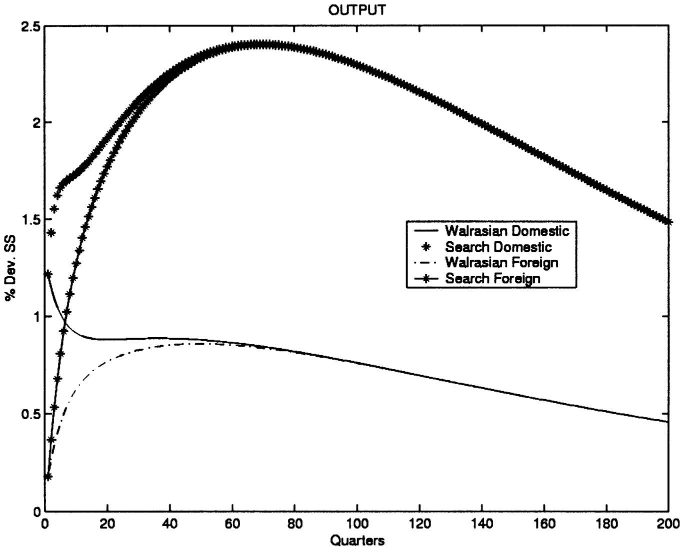

FIG. 10. Responses to a domestic shock (non-separable utility).

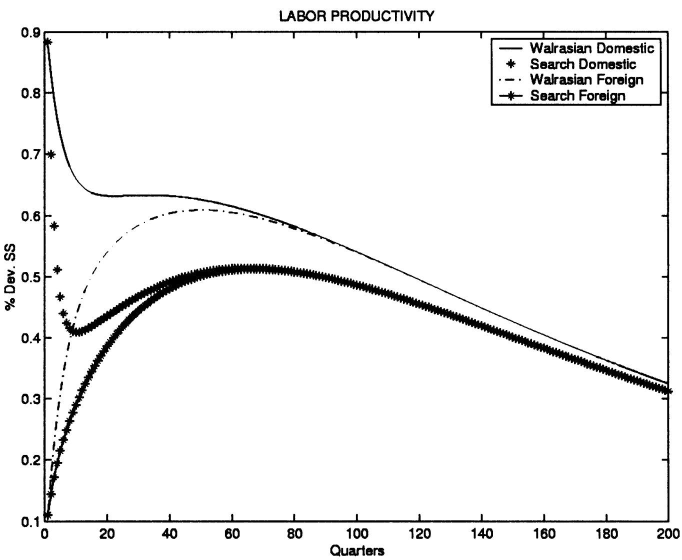

FIG. 11. Responses to a domestic shock (non-separable utility).

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